

ADVANCED MACHINE LEARNING · WORKSHEET 10

Feature Selection

Duplicate Removal · Variance & Correlation Filters · Wrapper Search · Validation Discipline

TEACHER COPY — Lecture 10

PART I: WHY FEATURE SELECTION?

SECTION 1 More Columns Do Not Always Mean More Information

▷ HOOK

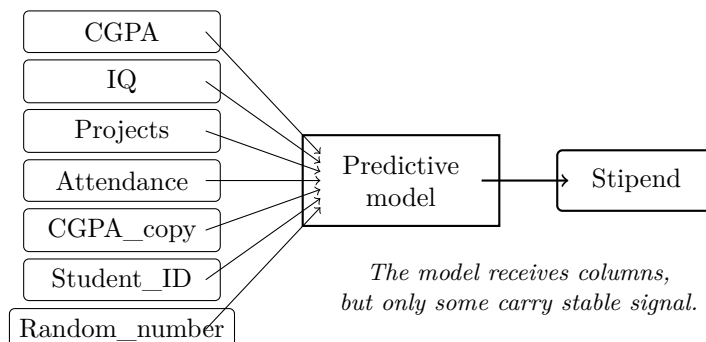
In Lecture 8, we deliberately created features such as x^2 so that a model could represent a curved pattern. In Lecture 9, we saw that extra flexibility can reduce bias but can also increase variance. We now ask the complementary question:

When a dataset already contains many columns, which columns deserve to remain?

Return to the internship-stipend problem. The target is a student's stipend, and the available candidate inputs are:

Feature	Why it may appear useful	Question to ask
CGPA	Academic performance may influence shortlisting.	Does it add stable predictive signal?
IQ score	May relate to problem-solving ability.	Does it help after other features are present?
Number of projects	May reflect practical experience.	Is its contribution linear or interaction-based?
Attendance	Could reflect discipline.	Is the relationship real or accidental in this sample?
CGPA_copy	Exactly repeats CGPA.	Does it contribute any new information?
Student_ID	Has large numerical variation.	Is it a genuine predictor or only an identifier?
Random_number	Changes from student to student.	Does variation alone make it relevant?

Visual: Seven candidate columns enter one prediction problem



Classroom question: Would the model necessarily become better if we keep all seven columns?

Expected answer: No. Duplicate, irrelevant, or unstable columns add no new signal and may increase variance, computation, and interpretability problems.

SECTION 2 Feature Selection: The Problem

A **feature** is an **input variable or column** used by a model to predict a target. **Feature selection** is the process of choosing a **smaller subset of relevant features** while preserving, or ideally improving, performance on unseen data.

★ KEY INSIGHT

“Most important” is not an absolute label. It depends on the **target, model, evaluation metric, sample, and the other features already present.**

2.1 Three kinds of feature behaviour

Type	Meaning	Running example
Relevant	Carries stable information that helps predict the target.	Number of projects may help predict stipend.
Redundant	Repeats information already carried by another feature.	CGPA_copy repeats CGPA.
Irrelevant	Varies but does not carry a stable relationship with the target.	Random_number may vary and still be useless.

◆ TAKEAWAY

Feature selection tries to remove **redundant** and **irrelevant** inputs without discarding genuinely useful signal.

2.3 Feature selection is not feature transformation

	Feature transformation	Feature selection
Core action	Create a new representation, such as x^2 from x .	Retain only some existing candidate columns.
Effect on candidate space	Expands or changes the representation.	Reduces the representation.
Can both appear in one pipeline?	Yes. Create meaningful candidates first, then select among them.	

SECTION 3 Why Feature Selection Can Improve Learning

3.1 Generalization and the bias–variance connection

More features increase a model’s flexibility. While useful features reduce bias, irrelevant features cause the model to fit random noise, increasing **variance** and hurting validation performance.

Classroom question: Training error decreases after ten features are added, but validation error increases. Did the new features help the model learn the stable pattern?

Expected answer: No. The model became more flexible on the training sample, but its generalization became worse. The pattern is consistent with unnecessary flexibility and increased variance.

3.2 Computation, Memory, and Interpretability

- **Efficiency:** Smaller matrices, faster training/prediction, lower memory use.
- **Explainability:** A model with fewer features is easier to explain.
- **Stability:** Removing redundant inputs reduces multicollinearity, making linear coefficients more stable.

SECTION 4 The Two Feature-Selection Families

When selecting features, we generally choose between two main approaches: **Filter methods** and **Wrapper methods**.

- **Filter methods** use simple statistical checks (like correlation) to screen features before any model is trained.
- **Wrapper methods** train the actual predictive model on different subsets of features to see which subset performs best.

We will explore both of these approaches in the following sections.

PRACTICE P1 · FEATURE TRIAGE AND FOUNDATIONS

- (a) For each feature, decide whether a simple filter should **Remove**, **Flag for investigation**, or **Keep for model-based evaluation**. Give one reason.

Feature	Observed property	Decision	Reason
CGPA_copy	Exactly equals CGPA in every row.	Remove	Exact duplicate; no new information.
Student_ID	Every value is unique; variance is high.	Flag / Remove	Likely an identifier, not generalizable signal.
Scholarship_flag	Only 3% of students have value 1.	Flag	Rare can still be predictive; inspect target relation.
Random_number	High variance and near-zero target correlation.	Remove / Flag	Strong removal candidate, but validate.
Project_depth ²	Highly correlated with project depth.	Keep / Flag	May be an intentional nonlinear feature.

- (b) **Choose all correct options.**

- A feature's usefulness can change when the target or model changes. **[T]**
- A high-variance column must be predictive. **[F]**
- Feature transformation and feature selection can both appear in one pipeline. **[T]**
- The smallest possible subset is always the best subset. **[F]**
- If two subsets have nearly equal cross-validated performance, the smaller subset is usually preferred. **[T]**

PART II: FILTER METHODS – SCREEN FEATURES BEFORE THE FINAL MODEL

SECTION 5 Filter Intuition and Duplicate Removal

▷ HOOK

A filter method acts like an **admission gate**. It evaluates a feature using a simple structural or statistical rule and decides whether that feature deserves to enter the modeling stage. The final predictive model is not repeatedly trained during this screening step.

Some filters do not use the target; others compare a feature with the target.

If two columns contain exactly the same value in every row, they provide the same information:

$$f_a = f_b \quad \text{for every observation.}$$

Keeping both does not create a second independent source of evidence. In the stipend example, CGPA and CGPA_copy are exact duplicates. For a linear model, retaining both also creates **perfect multicollinearity**, so their individual contributions cannot be uniquely separated.

Which copy should be kept?

- Keep the column with the clearer and more meaningful name.
- Prefer the column with fewer missing or corrupted values.
- Prefer the feature that is easier and cheaper to collect at prediction time.
- Check that the apparent duplicate is not a data-leakage column created after the target became known.

◆ TAKEAWAY

Duplicate removal is safe because it removes **repeated information**, not unique information.

SECTION 6 Variance Threshold

▷ HOOK

A feature that barely changes across observations cannot distinguish one observation from another. The simplest case is a constant column: if every student has the same value, that feature cannot explain why stipends differ.

For feature f_j , a variance measure is

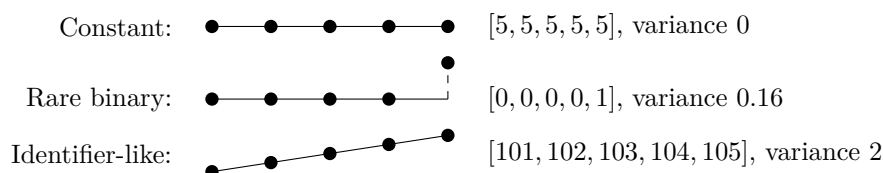
$$\text{Var}(f_j) = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2.$$

Choose a threshold τ . Remove the feature when

$$\text{Var}(f_j) < \tau.$$

The threshold is a **modeling choice**, not a universal constant.

Visual: What a variance filter sees



Feature values	Variance	What the filter sees	What must be remembered
[5, 5, 5, 5, 5]	0	No variation; remove.	A constant feature cannot separate observations.
[0, 0, 0, 0, 1]	0.16	Low variation; may be removed for a high τ .	A rare event can still be strongly predictive.
[101, 102, 103, 104, 105]	2	Moderate variation; may be kept.	An identifier can vary greatly and still be irrelevant.

Limitations of a variance threshold

- A high-variance feature may have no relationship with the target.
- A low-variance feature may be highly predictive, especially for a rare but important condition.
- The method ignores combinations and interactions among features.
- The threshold depends on units, scaling choices, and context.

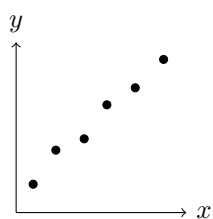
SECTION 7 Pearson Correlation Between Each Feature and the Target

▷ HOOK

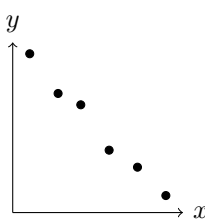
Variance tells us whether a feature changes. It does not tell us whether the feature changes **together with the target**. For a numerical feature and numerical target, Pearson correlation measures the strength and direction of a **linear** relationship.

$$r_{jY} = \frac{\sum_i (x_{ij} - \bar{x}_j)(y_i - \bar{y})}{\sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 \sum_i (y_i - \bar{y})^2}}.$$

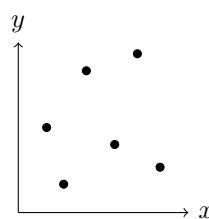
- r close to +1: strong positive linear relationship.
- r close to -1: strong negative linear relationship.
- r close to 0: weak linear relationship, **not necessarily no relationship**.



Strong Positive
($r \approx +1$)



Strong Negative
($r \approx -1$)



Weak/No Linear
($r \approx 0$)

An illustrative rule is

Remove f_j when $|r_{jY}| < 0.30$.

The value 0.30 is only an illustration; the threshold must be validated through model performance.

Feature	Illustrative r with stipend	Decision under $ r \geq 0.30$	Caution
CGPA	0.68	Keep	Strong linear association in this sample.
IQ score	0.18	Remove	Could still help jointly or nonlinearly.
Number of projects	0.57	Keep	Useful linear association.
Attendance	0.09	Remove	Threshold decision must be validated.
Student_ID	-0.02	Remove	Large variance did not imply relevance.

7.1 The nonlinear counterexample

Let $y = x^2$ for symmetric values of x :

x	-2	-1	0	1	2
$y = x^2$	4	1	0	1	4

1. Calculate $\bar{x}$ and $\bar{y}$:

$$\bar{x} = 0, \quad \bar{y} = 2.$$

2. Compute the numerator $\sum (x_i - \bar{x})(y_i - \bar{y})$:

$$\begin{aligned} \sum (x_i - 0)(y_i - 2) &= (-2)(2) + (-1)(-1) + (0)(-2) + (1)(-1) + (2)(2) \\ &= -4 + 1 + 0 - 1 + 4 \\ &= 0. \end{aligned}$$

3. Therefore, $r_{XY} = 0$.

A $|r| \geq 0.30$ filter would **remove x** , even though y is completely determined by x .

★ KEY INSIGHT

A feature can have zero Pearson correlation and still be perfectly predictive through a nonlinear relationship. Pearson correlation measures **linear association only**.

Limitations of target correlation

- It can miss nonlinear relationships.
- It evaluates one feature at a time and can miss useful interactions.
- It is sensitive to outliers.
- Its threshold is context-dependent.
- Correlation does not establish causation or future stability.

SECTION 8 Correlation Among Input Features

The previous calculation compared each feature with y . Correlation can also compare input features with one another to detect redundancy and reduce multicollinearity.

Suppose f_1 and f_2 have correlation 0.95. They carry very similar linear information. A practical rule is to keep one representative feature, but the choice should be deliberate.

How to choose which correlated feature to keep

- Prefer the feature with the stronger validated relationship with the target.
- Prefer the feature with less missingness and measurement noise.
- Prefer the feature that is easier to interpret and collect.
- Use domain knowledge before removing intentionally engineered terms.

△ WARNING – DO NOT APPLY PAIRWISE RULES BLINDLY

The features x and x^2 can be correlated, but both may be needed to represent curvature. A high input correlation does not automatically prove that one engineered feature should be removed. Check the final decision through validation performance.

8.1 Filter-method summary

Strengths	Limitations
Simple, fast, and scalable.	Usually ignores interactions among multiple features.
Can remove obvious problems before expensive modeling.	A statistical score may not match the needs of the final model.
Often model-independent and easy to explain.	Threshold choice can be arbitrary or unstable.
Useful as the first stage of a larger pipeline.	Univariate linear measures can miss nonlinear signal.

◆ TAKEAWAY

Filter methods are efficient, but they do not directly ask whether the chosen model performs better with a particular **combination** of features.

PRACTICE P2 · FILTER-METHOD DETECTIVE

(a) **Match the filter to the issue.**

	Observed issue		Most direct screen
1	Two columns are equal in every row.	A	Target correlation
2	A column is constant.	B	Input–input correlation
3	A numeric feature has weak linear association with numeric y .	C	Duplicate removal
4	Two predictors have correlation 0.97.	D	Variance threshold

Answers: $1 \rightarrow C$, $2 \rightarrow D$, $3 \rightarrow A$, $4 \rightarrow B$.

(b) **Find the mistake.** “Student_ID has very high variance, so a variance filter proves that it is useful for predicting stipend.”

Expected answer: Variance measures spread, not relevance. Student_ID may vary greatly while carrying no stable relationship with stipend. Use domain reasoning and validated predictive evidence.

(c) Scholarship_flag equals 1 for only 3% of students. Should it be removed solely because its variance is low? **[F]** Justify.

Expected answer: No. A rare binary feature can have low variance and still be strongly predictive. Its target relationship and validation contribution must be examined.

(d) Project_depth and Project_depth² are highly correlated. A student immediately removes the squared term. Is this decision justified? **[F]** Justify.

Expected answer: No. The squared term may be intentionally engineered to represent curvature. Pairwise correlation alone should not override the modeling purpose; compare validation performance.

PART III: WRAPPER METHODS – LET MODEL PERFORMANCE JUDGE THE SUBSET

SECTION 9 The Wrapper Idea and the Wrapper Loop

▷ HOOK

A feature can look weak by itself and become useful when paired with another feature. Conversely, two individually useful features may become redundant when used together. Wrapper methods evaluate **whole subsets** with the predictive model we actually plan to use.

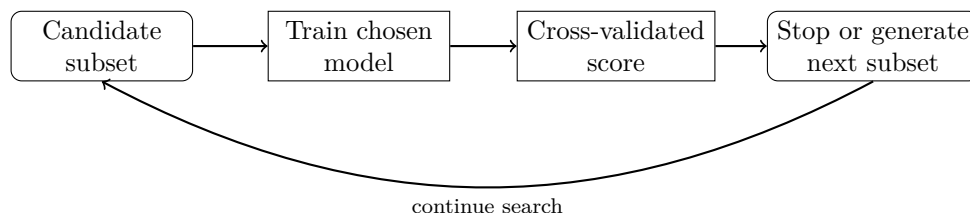
- Filters evaluate features one at a time, ignoring how they work together.
- Filters can miss useful interactions between features.
- A simple statistical score might not align with how your complex model learns.

★ KEY INSIGHT

A wrapper method “wraps” model training and validation around the feature-selection process.

9.1 The wrapper loop

1. Generate a candidate subset of features.
2. Train the chosen model on that subset.
3. Evaluate the subset using validation data or cross-validation.
4. Generate the next candidate subset according to the search strategy.
5. Stop when the search is complete or a stopping rule is met.



△ WARNING – USE CROSS-VALIDATED SCORES

When adding features, a model’s performance on the **training data** will almost always look better, even if the new features are useless. To truly evaluate a subset, you must use a **cross-validated score**. Cross-validation tests the model on data it hasn’t seen during training, giving a much more honest estimate of how well the features actually work.

SECTION 10 Exhaustive Feature Selection

Exhaustive search evaluates every nonempty subset and chooses the subset with the best validation score. Each of p features has two states: included or excluded. Therefore,

$$\text{Number of nonempty subsets} = 2^p - 1.$$

Candidate features p	Nonempty subsets $2^p - 1$
4	15
20	1,048,575
30	1,073,741,823

Why exhaustive search becomes impractical

- **Advantage:** if every subset is evaluated correctly, the best subset in the search space can be found.
- **Disadvantage:** computational cost grows exponentially.
- **Disadvantage:** testing many subsets can overfit the validation procedure and select a chance winner.
- **Disadvantage:** the result depends on the chosen model and evaluation metric.

SECTION 11 Backward Elimination**▷ HOOK**

Backward elimination starts with all features. At each stage, remove each remaining feature one at a time, evaluate the resulting subsets, and keep the best candidate for the next stage.

Treat every score below as cross-validated R^2 .

11.1 Start with all four features

Current subset	Cross-validated R^2
$\{f_1, f_2, f_3, f_4\}$	0.89

11.2 Remove one feature from the four-feature set

Candidate subset	Cross-validated R^2
$\{f_2, f_3, f_4\}$	0.81
$\{f_1, f_3, f_4\}$	0.71
$\{f_1, f_2, f_4\}$	0.91
$\{f_1, f_2, f_3\}$	0.65

Continue with $\{f_1, f_2, f_4\}$; the removed feature is f_3 . Notice that validation performance improved from 0.89 to 0.91.

11.3 Repeat on $\{f_1, f_2, f_4\}$

Candidate subset	Cross-validated R^2
$\{f_2, f_4\}$	0.79
$\{f_1, f_4\}$	0.81
$\{f_1, f_2\}$	0.83

Continue with $\{f_1, f_2\}$; remove f_4 .

11.4 Compare the remaining single-feature models

Candidate subset	Cross-validated R^2
$\{f_2\}$	0.63
$\{f_1\}$	0.53

The path therefore keeps $\{f_2\}$ at the one-feature stage.

11.5 Choose the best subset encountered along the path

Best subset at that size	Cross-validated R^2
$\{f_1, f_2, f_3, f_4\}$	0.89
$\{f_1, f_2, f_4\}$	0.91
$\{f_1, f_2\}$	0.83
$\{f_2\}$	0.63

The selected subset is $\{f_1, f_2, f_4\}$ because it achieved the highest cross-validated R^2 ; it is not selected merely because it is the final or smallest subset.

11.6 Computational advantage and greedy limitation

For $p = 4$, backward elimination fits

$$1 + 4 + 3 + 2 = 10$$

candidate models instead of all 15 nonempty subsets. In general,

$$1 + p + (p - 1) + \cdots + 2 = \frac{p(p + 1)}{2}.$$

This grows quadratically rather than exponentially, although every score may still require several cross-validation fits.

⚠ WARNING – GREEDY SEARCH

Backward elimination makes a locally best decision at each stage. Once a feature is removed, it is never reconsidered. The method can therefore miss a globally best combination that contains an earlier discarded feature.

SECTION 12 Sequential Forward Selection

▷ HOOK

Forward selection begins with no input features. For regression, the baseline model predicts the mean target $\bar{y}$. The method adds the feature that gives the largest validation improvement, fixes that choice, and repeats.

12.1 Baseline model

$$\text{No input feature} \implies \hat{y} = \bar{y}.$$

12.2 Add one feature

Candidate subset	Cross-validated R^2
$\{f_1\}$	0.63
$\{f_2\}$	0.51
$\{f_3\}$	0.43
$\{f_4\}$	0.49

Select f_1 .

12.3 Add one feature to $\{f_1\}$

Candidate subset	Cross-validated R^2
$\{f_1, f_2\}$	0.63
$\{f_1, f_3\}$	0.71
$\{f_1, f_4\}$	0.80

Select $\{f_1, f_4\}$.

12.4 Add one feature to $\{f_1, f_4\}$

Candidate subset	Cross-validated R^2
$\{f_1, f_4, f_2\}$	0.81
$\{f_1, f_4, f_3\}$	0.85

Select $\{f_1, f_4, f_3\}$.

12.5 Try the final feature

Candidate subset	Cross-validated R^2
$\{f_1, f_4, f_3, f_2\}$	0.83

The all-feature model scores 0.83, below the best three-feature score 0.85. The best subset encountered is therefore $\{f_1, f_4, f_3\}$.

◆ TAKEAWAY

Forward selection does not automatically keep adding features until all are used. Stop when an additional feature no longer gives a meaningful validation improvement, or choose the best subset along the entire forward path.

12.6 When forward or backward selection is more natural

- Use **forward selection** when p is large and the desired final subset is small; for example, keep about 10 of 100 features.
- Use **backward elimination** when the model can fit all p features and the desired final subset retains most of them; for example, keep about 90 of 100 features.
- Both are greedy and model-specific, so they can select different subsets.

12.7 Wrapper-method summary

Strengths	Limitations
Directly optimizes performance for the chosen model and metric.	Computationally expensive, especially with cross-validation.
Can capture useful combinations and interactions that filters miss.	Can overfit the feature-search procedure.
Produces a subset tailored to the final modeling goal.	Selected features may not transfer to a different model.
Sequential methods are much cheaper than exhaustive search.	Greedy methods can miss the globally best subset.

PRACTICE P3 · WRAPPER SEARCH AND REASONING

- (a) The current backward-elimination subset is $\{f_1, f_2, f_4\}$. Removing one feature gives:

Candidate subset	Cross-validated R^2
$\{f_2, f_4\}$	0.79
$\{f_1, f_4\}$	0.81
$\{f_1, f_2\}$	0.83

- (i) Which subset continues? $\{f_1, f_2\}$
 (ii) Which feature is removed? f_4
 (iii) Why can the procedure still miss the globally best subset?

Expected answer: Backward elimination is greedy. Once a feature is removed, it is never reconsidered, so the search may never evaluate a globally strong combination that contains an earlier discarded feature.

- (b) For $p = 6$, compare the number of candidate models:

$$\text{Exhaustive: } 2^6 - 1 = \mathbf{63}, \quad \text{Backward path: } \frac{6(6+1)}{2} = \mathbf{21}.$$

State the growth pattern of each method.

Expected answer: Exhaustive search grows exponentially; the backward path grows quadratically.

- (c) Arrange the wrapper loop in the correct order. Write the letters in sequence.

- A. Evaluate with validation data or cross-validation.
 B. Generate the next candidate subset.
 C. Train the chosen model on the candidate subset.
 D. Stop when the search or stopping rule is complete.
 E. Generate an initial candidate subset.

Correct order: **E → C → A → B → D**.

- (d) **Assertion–Reason.**

A: If exhaustive search evaluates every nonempty subset correctly, it can find the best subset in that search space.

R: Exhaustive search evaluates all $2^p - 1$ nonempty subsets.

- (i) Both A and R are true, and R is the correct explanation of A. **[T]**
 (ii) Both A and R are true, but R is not the correct explanation of A. **[F]**
 (iii) A is true but R is false. **[F]**
 (iv) A is false but R is true. **[F]**

- (e) Match the search direction to the goal.

Goal	More natural method
From 100 features, keep about 10.	Forward selection
From 100 features, retain about 90.	Backward elimination

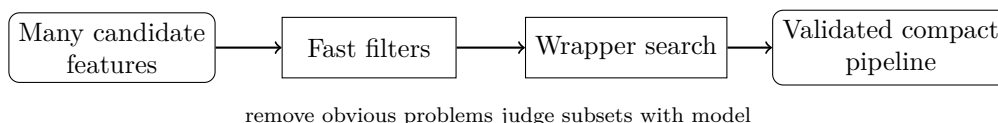
PART IV: CHOOSING A METHOD AND VALIDATING THE DECISION

SECTION 13 Filter and Wrapper Methods Compared Directly

Criterion	Filter	Wrapper
Uses final-model performance directly?	Usually no	Yes
Captures feature interactions?	Usually no	Yes, if the chosen model can represent them
Model-specific?	Usually no	Yes
Computational cost	Low	High
Best role	Fast screening and preprocessing	Careful subset search for a chosen model

13.1 Practical method-choice rules

- Start with filters when p is very large or when obvious duplicates and near-constant columns exist.
- Use a wrapper when the feature count is manageable, model-specific performance matters, and computational budget is available.
- Combine methods when useful: apply simple filters first, then run a wrapper on the reduced set.
- Always compare the selected-feature pipeline with an all-feature baseline using the same validation procedure.



SECTION 14 Validation Discipline: Avoid the Leakage Trap

▷ HOOK

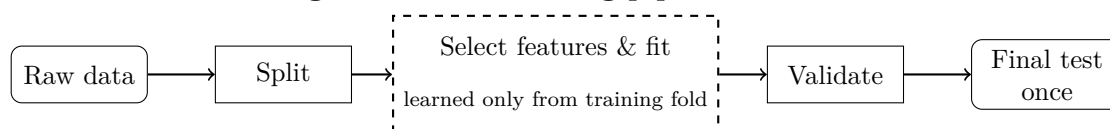
Feature selection is part of teaching your model. It is **not** a harmless step you can do before properly splitting your data.

△ WARNING – THE LEAKAGE TRAP

Wrong: Selecting your features using the entire dataset, and *then* splitting the data into training and validation sets. If you do this, your model has already “peeked” at the validation data during the feature selection step. This is called data leakage, and it will make your performance look falsely optimistic.

Right: Inside every cross-validation fold, perform feature selection using **only** the training portion of that fold. Then, evaluate it on the untouched validation portion.

Visual: Selection belongs inside the training pipeline



The final test set should never be used to choose which features to keep. It is used exactly once at the very end to estimate how well your final pipeline will perform in the real world.

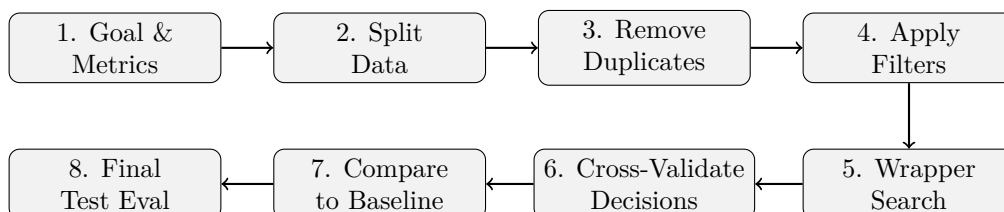
SECTION 15 The Selected Subset Is Not Universal

A selected feature set is supported by a particular experimental design; it is not a permanent truth.

- Change the **target**, and relevant features can change.
- Change the **model**, and the useful subset can change.
- Change the **evaluation metric**, and the preferred subset can change.
- Change the **sample**, and unstable selection procedures may choose different features.

Therefore, report the selected features together with the dataset, target, model, metric, and validation design that supported them.

SECTION 16 End-to-End Feature-Selection Workflow



Step 1: Define the target, evaluation metric, and final modeling goal.

Step 2: Create the train–validation–test structure before learning selection rules.

Step 3: Remove exact duplicates and structurally unusable columns.

Step 4: Use simple filters for obvious low-information or redundant inputs, while checking their limitations.

Step 5: Use a wrapper when model-specific subset evaluation is needed and computationally feasible.

Step 6: Choose thresholds, feature counts, and stopping rules inside cross-validation.

Step 7: Compare with the all-feature baseline; prefer the simpler model only when generalization is maintained or improved.

Step 8: Evaluate once on the untouched test set and report the final feature set with the method used to obtain it.

◆ TAKEAWAY

Feature selection is successful only when the smaller representation preserves useful signal and improves the reliability, efficiency, or interpretability of the complete predictive pipeline.

16.1 Closing synthesis

1. More features can add signal, but they can also add redundancy, noise, computation, and variance.
2. Filter methods screen quickly using structural or statistical rules, but usually miss interactions.
3. Wrapper methods judge candidate subsets using the chosen model, but repeated fitting is expensive and greedy searches can miss the global optimum.
4. Selection must occur inside validation; the test set remains untouched until every decision is fixed.

PRACTICE P4 · COMPREHENSIVE CHALLENGE

(a) **Diagnose the pipeline.** A student performs these steps:

- (1) Compute correlations using the entire dataset.
- (2) Keep the top five features.

- (3) Run five-fold cross-validation using those five columns.
- (4) Report the mean validation score.

Is the reported score trustworthy? **[F]** Explain the problem and correct the procedure.

Expected answer: The score is optimistic because the validation folds influenced feature selection. Recompute the correlation filter separately inside each training fold, apply that fitted selector to its validation fold, fix all choices through cross-validation, and use the untouched test set only once at the end.

Marking hint: Full credit requires naming leakage and placing feature selection inside each training fold.

- (b) Choose the most appropriate starting strategy for each situation.

Situation	Starting strategy
Thousands of columns contain exact duplicates and near-constant fields.	Simple filters first
A manageable feature set may contain model-specific interactions, and sufficient computation is available.	Wrapper method
Obvious problems can be removed cheaply, after which a smaller model-aware search is feasible.	Filter, then wrapper

- (c) **True or False.** Justify each answer briefly.

Statement	T/F	Justification
The smallest possible feature set is always best.	[F]	It may discard useful signal; choose by validated performance.
A low-variance feature is always useless.	[F]	Rare binary features can still be predictive.
Wrapper methods usually cost more because they repeatedly fit and evaluate models.	[T]	Model fitting occurs for many candidate subsets and folds.
Two different models can legitimately select different features from the same dataset.	[T]	Usefulness is model-specific and interaction-dependent.
The test set may be used to choose the correlation threshold.	[F]	The test set is used once after all choices are fixed.

- (d) Arrange the following stages into a trustworthy sequence.

- A. Evaluate once on the untouched test set.
- B. Split the data before learning selection rules.
- C. Define the target, metric, and modeling goal.
- D. Fit preprocessing, selection, and the model inside cross-validation.
- E. Compare with the all-feature baseline and fix all choices.

Correct order: **C → B → D → E → A.**

- (e) Two subsets have nearly the same cross-validated performance. One uses 18 features and the other uses 7. Which would you normally prefer, and what could justify the larger subset?

Expected answer: Normally prefer the 7-feature subset because it gives a simpler representation with nearly unchanged generalization. Retain the larger subset only when domain knowledge, measurement reliability, operational requirements, or a meaningful validated advantage provides a strong reason.

(f) Why can the selected features change when the target, model, metric, or sample changes?

Expected answer: Feature usefulness is conditional on the prediction problem and validation design. Different targets, model capabilities, optimization criteria, or samples can change which signals are useful, redundant, or stable.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Explain feature selection as choosing a subset of existing inputs.
- ☐ Distinguish relevant, redundant, and irrelevant features.
- ☐ Connect unnecessary features with variance, sparse high-dimensional coverage, computation, and interpretability.
- ☐ Distinguish feature selection from feature transformation.
- ☐ Apply duplicate removal, variance threshold, target correlation, and input-correlation screens with appropriate cautions.
- ☐ Explain why zero Pearson correlation does not imply zero predictive value.
- ☐ Calculate the number of exhaustive subsets and trace backward and forward sequential searches.
- ☐ Compare filter and wrapper methods by cost, interaction-awareness, and model dependence.
- ☐ Place selection correctly inside cross-validation and protect the final test set.
- ☐ Report a selected subset as conditional on the target, model, metric, sample, and validation design.

ANSWER KEY AND MARKING GUIDE

P1. (a) CGPA_copy: Remove; Student_ID: Flag/remove as an identifier; Scholarship_flag: Flag, because rare can still be predictive; Random_number: strong removal candidate, but validate; Project_depth²: keep/flag because it may encode curvature. (b) T, F, T, F, T.

P2. (a) $1 \rightarrow C$, $2 \rightarrow D$, $3 \rightarrow A$, $4 \rightarrow B$. (b) Variance measures spread, not relevance. (c) False; a rare binary feature can be predictive. (d) False; x^2 may be needed for curvature.

P3. (a) Keep $\{f_1, f_2\}$, remove f_4 ; the search is greedy and never reconsiders removed features. (b) Exhaustive = 63; backward path = 21; exponential versus quadratic growth. (c) $E \rightarrow C \rightarrow A \rightarrow B \rightarrow D$. (d) Option (i). (e) Forward selection; backward elimination.

P4. (a) False; the procedure leaks validation information. Fit the selector separately inside each training fold and reserve the test set for one final evaluation. (b) Filters first; wrapper; filter then wrapper. (c) F, F, T, T, F. (d) $C \rightarrow B \rightarrow D \rightarrow E \rightarrow A$. (e) Normally prefer the 7-feature subset unless domain or operational evidence justifies the larger set. (f) The selected subset is conditional on the target, model, metric, sample, and validation design.